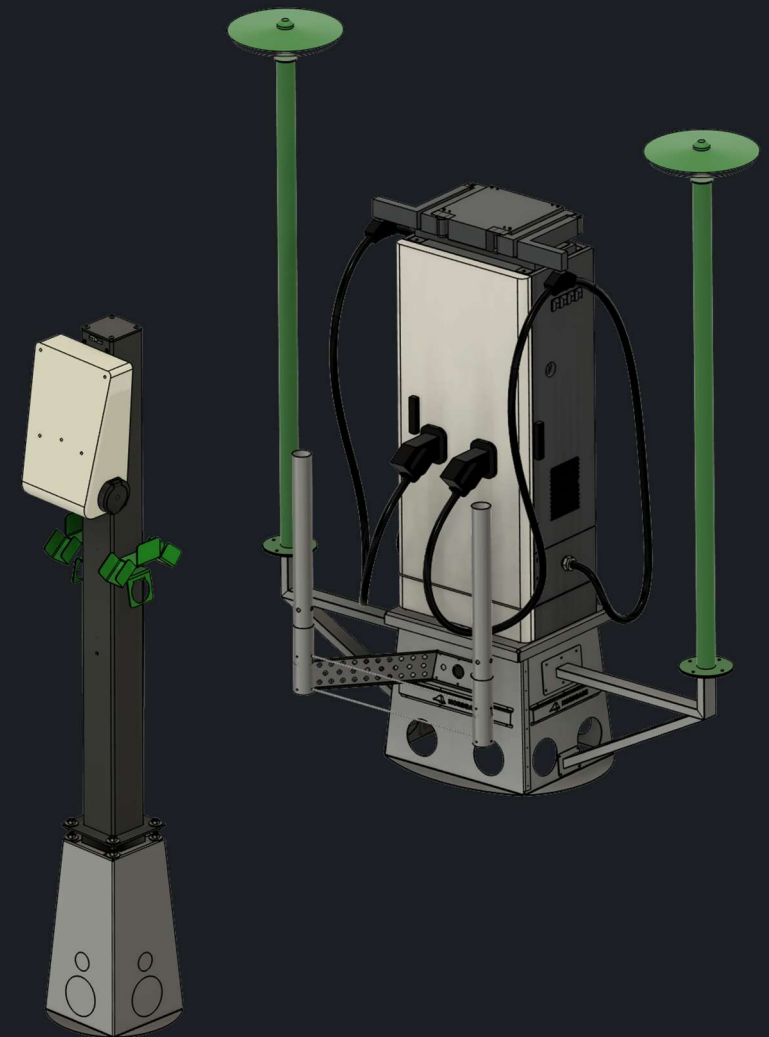




Installation Manual

NordBase AC & Bollard Foundation





INTRODUCTION

NordBase Foundation System

NordBase is a concrete-free steel foundation system for EV charger and bollard installations. Bolt-together assembly, no cure time, no rebar — ready to load the moment it's backfilled.

NORDINFRA

Rev. 1.1 — 2026-08-07

INTRODUCTION

NordBase AC & Bollard Foundation

NordBase is a modular steel foundation for EV charging equipment and bollards — engineered to replace poured concrete. Made from ZAM-coated steel using kerf-fold assembly, it's bolt-together rather than cast: no cure time, no on-site welding. One foundation platform, one shared bolt pattern, for both AC chargers and bollards.

TOOLS AND MATERIALS REQUIRED



Wrench / torque tool



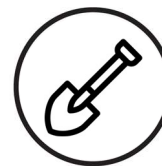
PPE



Leveling tool



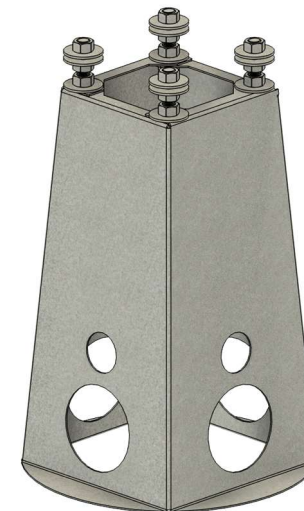
Compactor



Shovel

SCOPE OF DELIVERY

- Foundation body — pre-assembled steel shell with factory-installed stud bolt kit on the top plate
- Hardware kit — nuts, washers, and stud connection, factory-installed — no loose hardware bags to lose on site
- Adapter plate or bollard base plate — model-specific, bolts directly to the foundation's top plate pattern





OVERVIEW

NordBase Foundation System

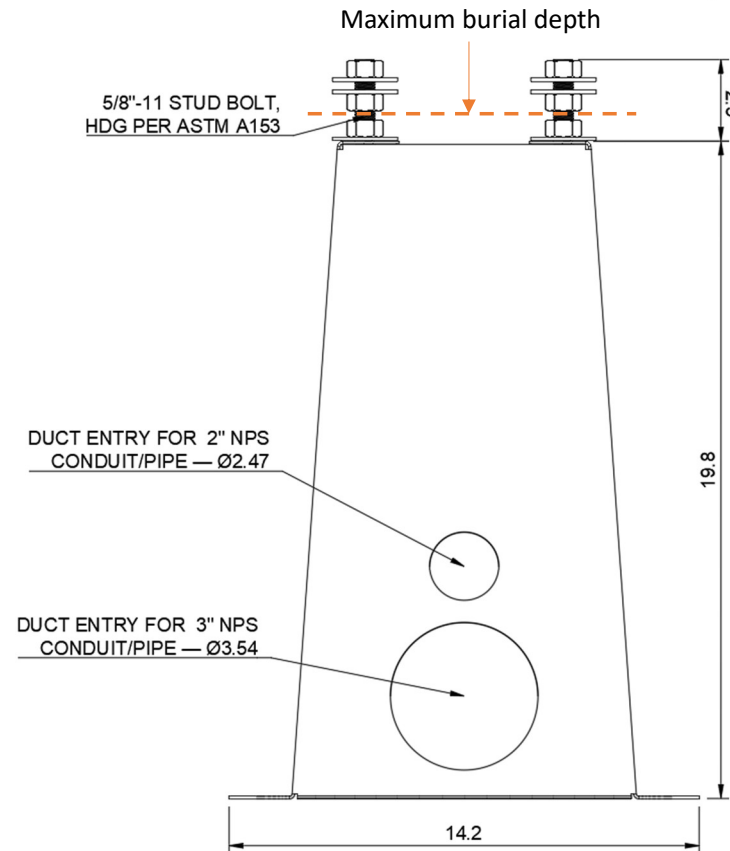
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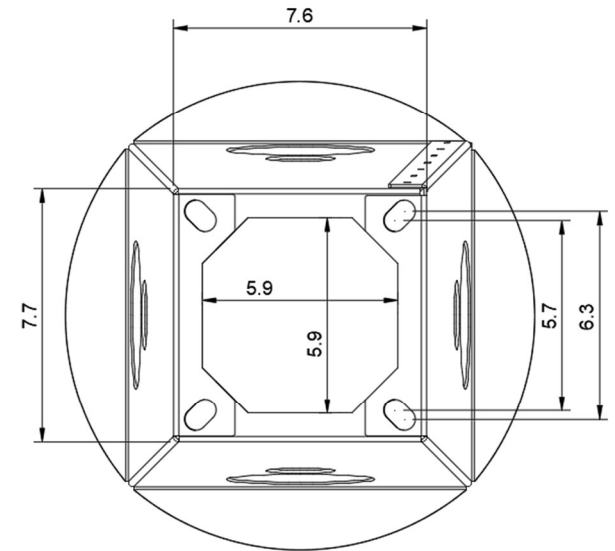
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PRODUCT OVERVIEW

NordBase AC & Bollard Foundation



ALL DIMENSIONS IN INCHES [±0.06"] UNLESS OTHERWISE SPECIFIED. DO NOT SCALE DRAWING.





INSTALLATION

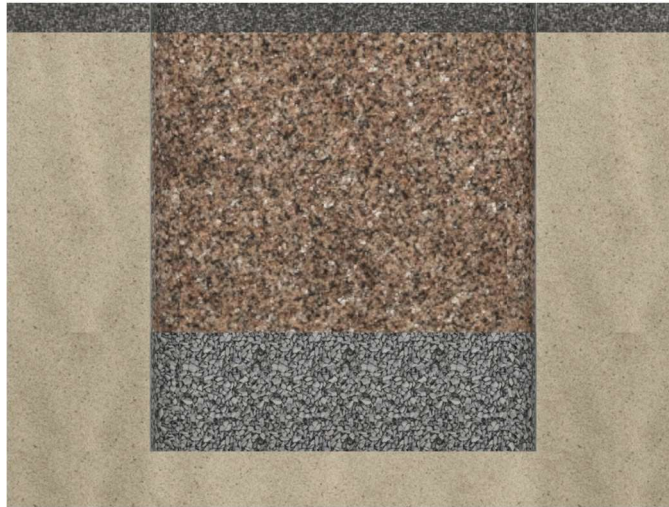
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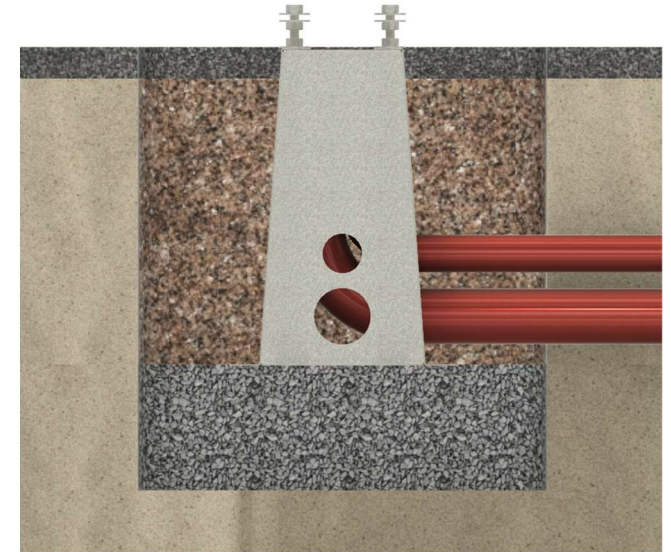
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1 Prepare the Trench



Step 1.1: Excavate the Trench

Excavate to the required depth and width for the foundation being installed. Lay a 8" layer of compacted base course / crushed stone at the bottom of the trench before placing the foundation.



Step 1.2: Position the Foundation

Place the foundation on the compacted base course. Align conduit openings with the planned cable routing, then route conduit through the entry point before backfilling.



Call 811 (or your local utility notification center) at least 2-3 business days before excavating. Confirm the area is free of underground utilities before digging.



INSTALLATION

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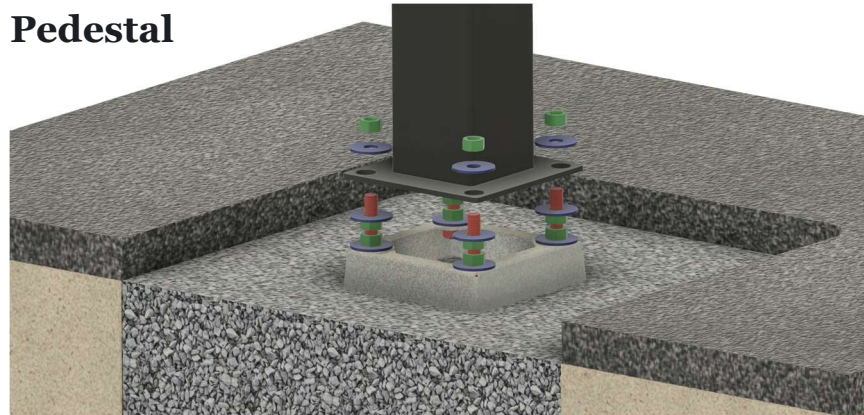
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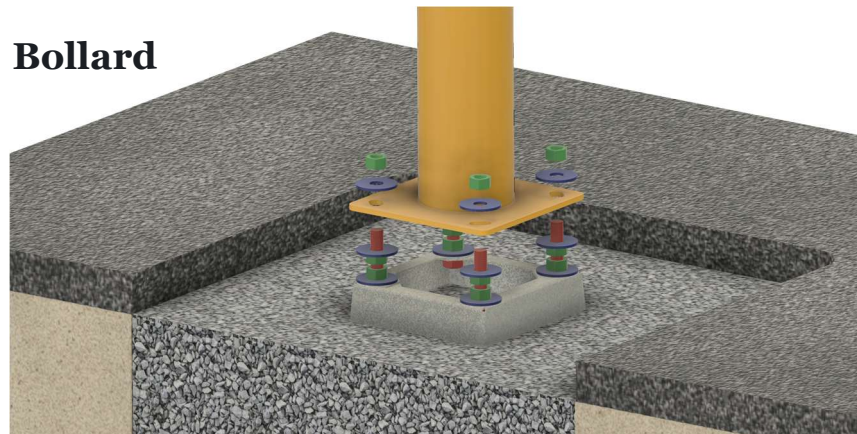
2 Backfill and Compact



Pedestal



Bollard



REFERENCE POINTS

Step 2.1: Stud Bolts

Factory-installed in the foundation's top plate. No under-access is needed or possible once the foundation is set — do not attempt to remove or reposition in the field.

Step 2.2: Bracing

Before backfilling, temporarily brace or support the foundation to prevent movement during compaction. Confirm the foundation is both level and plumb (vertically straight) — check both before proceeding.

Step 2.3: Surround Backfill

Crushed stone compacted in layers within the 200mm (8") zone around the foundation. Provides free drainage and lateral stability.

Step 2.4: Mounting Hardware

Nuts and washers secure the pedestal or bollard base plate directly to the foundation's bolt pattern.

Instructions

Backfill in 150-200mm (6-8") lifts, compacting thoroughly after each layer. Re-check the foundation with a spirit level after each lift. Confirm the top plate is level and flush with finished grade before final tightening.



REFERENCE

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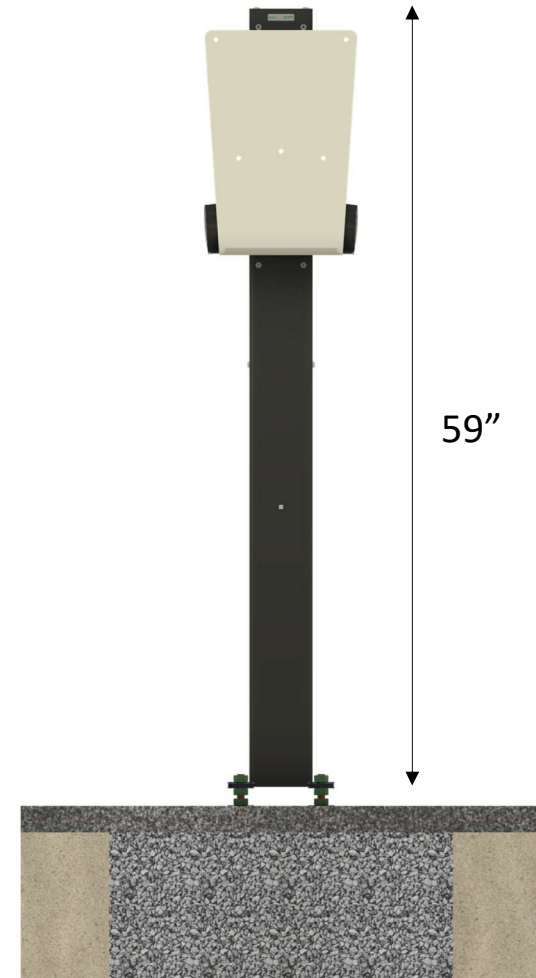
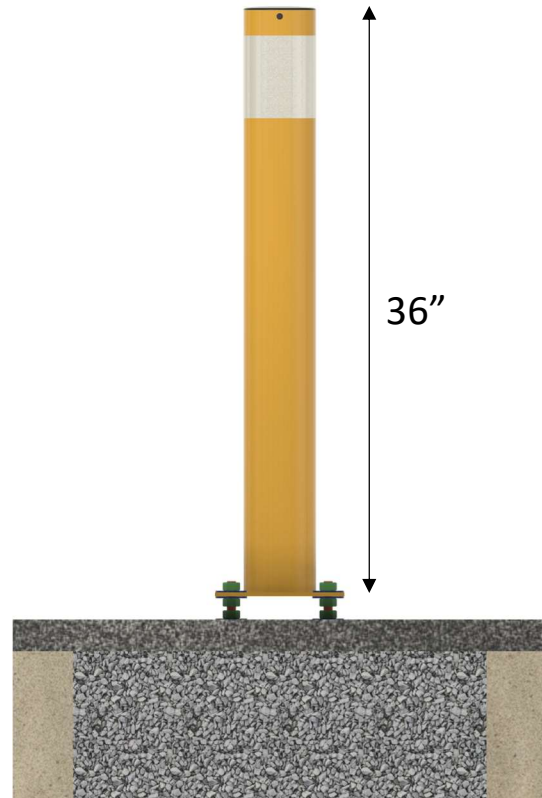
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TECHNICAL REFERENCE

Installed product



Completed installation — pedestal (left) and bollard (right), foundation flush with finished grade.

Note: To cover stud bolts — add our Bollard Base Cover — Page 6



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PRODUCT LINEUP

Accessories

Compatible hardware and covers for foundation and bollard installations.



Bollard

Sch. 10 (standard) / Sch. 40 (heavy-duty)

36" x 4.5" OD, ASTM A500 Gr. B
Powder coat / HDG + powder coat



Bollard Base Cover

14 GA carbon steel, spun

Ø11.4", Ø4.6", 2,6"
Powder coat, OSHA Safety Yellow



Bolt Kit

Factory-installed stud bolts

5/8"-11, HDG per ASTM A153
No loose hardware on site

Bolt kit uses the foundation's factory-installed top plate pattern. Base cover slides over the bollard pipe to conceal mounting hardware.



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REFERENCE

General Recommendations

FINAL INSPECTION CHECKLIST

Before proceeding to pole or bollard mounting, confirm:

- ☒ Backfill material is properly compacted in all layers
- ☒ No voids or gaps remain around the foundation
- ☒ Top plate is flush with finished grade and aligned with the project layout
- ☒ Foundation is level and plumb
- ☒ Foundation is ready for hardware installation

ROW ALIGNMENT

For multiple installations in a row (e.g. parking lot bollards or charger banks), align each foundation's bolt pattern parallel to adjacent units and to any marked reference lines before backfilling.

SITE CONDITIONS

If excavation encounters bedrock, a high water table, or other unexpected ground conditions, consult a geotechnical engineer before proceeding. Standard installation depths assume typical soil conditions as outlined in this manual.

STORAGE BEFORE INSTALLATION

Store foundations on a flat, dry surface prior to installation. Avoid prolonged ground contact in standing water. Factory-installed hardware should remain covered or protected from debris until final assembly.

This manual provides general installation guidance based on standard practice and is not a substitute for project-specific engineering. The installing contractor is solely responsible for excavation dimensions, bedding preparation, foundation orientation, backfill material and compaction, conduit installation, and correct equipment mounting.



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TECHNICAL NOTES

Material & Compliance

COATING & CORROSION

NordBase foundations are fabricated from SS Grade 33 steel with a ZM115 (Zinc-Aluminum-Magnesium) coating. The ZM115 coating provides self-healing corrosion protection at cut edges — the zinc-aluminum-magnesium alloy sacrificially protects exposed steel even where the coating has been cut or scratched, unlike standard galvanizing. Minor surface rust may appear over time; this is cosmetic only and will not spread or affect structural performance.

Steel is domestically melted and manufactured in the USA, supporting Buy America, Buy American (BABA) requirements for federally funded infrastructure projects, including NEVI-funded EV charging installations.

APPLICABLE STANDARDS

- ASCE 7-22
- IBC 2021
- ASTM A1046
- NEC — National Electrical Code
- ASTM A500

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NORDBASE

Safety Information



Personal Safety

PPE requirement text — gloves, safety glasses, steel-toe boots.



Proper Handling

Handling guidance text for tools and components.



Public Safety

Guidance on protecting the public — bolt caps, barriers.



Utility Locates

Call 811 (or your local utility notification center) at least 2–3 business days before excavating. Have all underground utilities marked before digging.



Pennsylvania USA



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*This manual is a recommendation based on standard practice.
Final geotechnical suitability must be verified by the installer.*

